

(b) Using convolution theorem, evaluate $L^{-1} \left\{ \frac{1}{(s^2 + a^2)^2} \right\}$.

(06)

IET, DAVV, Indore
B.E. II (Civil)
Class Test III (Nov-2023)
3AVRC1: Applied Mathematics-III

Time: 70 min

Max Marks: 20

Note: Attempt any FOUR questions. All questions carry equal marks.

Q.1. Evaluate $\int_0^6 \frac{dx}{1+x^2}$ by using (i) Trapezoidal rule (ii) Simpson's 1/3 rule (iii) Weddle's Rules. (05)

Q.2. Calculate the first & second derivatives of the function tabulated below, at the point $x = 1.1$ (05)

x	1.0	1.2	1.4	1.6	1.8	2.0
y	0	0.128	0.544	1.296	2.432	4.000

Q.3 Find the Fourier series of the function $F(x) = x^2$, $-\pi$ to π . (05)

Q.4 Find the Fourier Transform of (05)

$$F(x) = 1 - x^2, |x| \leq 1$$

$$0, |x| \geq 1$$

Hence Evaluate $\int_0^\infty \frac{x \cos x - \sin x}{x^3} \cos \frac{x}{2} dx$.

Q.5. Define Sampling and its types. (05)
